Mixed Reality Conversational Support for Older Adults

4ZP6-G4

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Repo: https://github.com/Macstone-25/Mixed-Reality

Table of Contents

Table of Contents 1

Contributions 2

Software Requirements Specification 2

- Purpose of the Project 2
- Stakeholders 3
- Mandated Constraints 3
- Naming Conventions and Terminology 4
- Relevant Facts and Assumptions 4
- Functional Requirements 4
- Data and Metrics 5
- Non-functional Requirements 7
  - Look and Feel Requirements 7
  - Usability and Humanity Requirements 7
  - Performance and Speed Requirements 7
  - Operational and Environmental Requirements 8
  - Security and Privacy 8
- Risks and Issues 8
- User Documentation and Training 9

Project Development Plan 10

- Team Communication Plan 10
- Team Member Roles 10
- Workflow Plan 11
- Proof of Concept Demonstration Plan 12
- Technology 13
- Project Scheduling (Gantt chart) 14

Contributions

Name	Contributions
William Clubine	Purpose of the Project + Mandated Constraints + User Documentation + Team Member Roles + Workflow Plan
Gregory Archer	Functional Requirements + Stakeholders + Relevant FAA
Jasiri Wa-Kyendo	Data and Metrics + Security and Privacy + Proof of Concept Plan
Nicholas Zajkeskovic	Look & Feel + Performance & Speed Requirements + Technology
Mayowa Adesanya	Usability and Humanity Requirements + Operational and Environmental Requirements + Project Scheduling
Khach Kehyayan	Functional Requirements + Risks and Issues + Team Communication Plan + Project Scheduling

Software Requirements Specification

Purpose of the Project

Background of the Project Effort

This project is part of an ongoing research initiative aimed at designing accessible communication technologies to support social participation among older adults. It seeks to enhance engagement, satisfaction, and accessibility through the integration of mixed reality and artificial intelligence. Aging populations often face challenges such as cognitive decline, reduced mobility, and sensory impairments that can limit social interaction and independence.

We propose a mixed reality application for the Apple Vision Pro that will support older adults during real-time conversations by providing adaptive visual and auditory cues powered by speech recognition, large language models (LLMs), and behavioral signal analysis. These features will help sustain attention, encourage participation, and capture meaningful data for research while maintaining user privacy and ethical data practices.

Beyond immediate user interaction, the project also aims to contribute to broader societal outcomes, such as reducing loneliness and digital exclusion, promoting equitable access to emerging technologies, and upholding transparency and accountability in AI systems.

Goals of the Project

The project's goals have two main pillars: supporting older adults in having more engaging and accessible conversations, and creating a reliable platform for researchers to collect, analyze, and refine interaction data. From these pillars, five main goals emerge:

#	Goal	Measure of Success
1	Accurately detect when an older adult is disengaging or struggling to follow a conversation.	Achieving high accuracy correctly identifying events.
2	Enhance older adults' ability to have meaningful conversations with real time auditory/visual prompts.	Reduced conversational stall rate.
3	Increase older adults' confidence and comfort in technology-mediated social interactions.	High percentage of prompts accepted by users.
4	Enable controlled research through configurable parameters and comprehensive data collection.	Researcher feedback and scenarios recorded.
5	Protect participant privacy and ensure ethical data handling throughout all stages of research.	Verified anonymization and ethics board compliance.

These goals ensure that the project not only delivers measurable improvements in cognitive and social engagement but also advances the broader objective of creating inclusive, ethical, and human-centered technologies for older adults.

Stakeholders

Primary Stakeholders

The primary stakeholders for the *Mixed Reality Conversational Support for Older Adults* project are the individuals and groups as follows:

- Dr. Melina Head, Dr. Aydin Farrokhi, and DeGroote School of Business Researchers;
- Professor Mehdi Moradi and the COMPSCI 4ZP6 Administrators;
- G. Archer, J. Wa-Kyendo, K. Kehyayan, M. Adesanya, N. Zajkeskovic, and W. Clubine

Secondary Stakeholders

The secondary stakeholders for the *Mixed Reality Conversational Support for Older Adults* project are the individuals and groups as follows:

- Apple Development Team (visionOS, Vision Pro);
- OpenAI (ChatGPT integration);
- Institutional Review Board (Research Ethics Board)

Mandated Constraints

The primary constraint for this project is the requirement to develop exclusively for the Apple Vision Pro, as this is the device selected and made available to us by the faculty supervisor. Development must therefore conform to Apple's visionOS ecosystem and associated SDK limitations. Other mixed reality platforms (such as Meta Quest or HoloLens) cannot be used. Additionally, the Apple Vision Pro imposes certain restrictions

on accessing raw camera feeds and lower-level system data without an enterprise license. While these limitations are not expected to impede the core functionality of the application, they define the technical boundaries within which the team must operate.

Naming Conventions and Terminology

Term	Definition
Vision Pro	Apple's flagship augmented / virtual reality device
visionOS	The operating system that powers Apple Vision Pro devices.
Prompt Generation	The generation of short conversational prompts shown to get a user back on track in a conversation when they become lost
Event Trigger Engine	The portion of the software responsible for analyzing the live transcript and determining if users could use a prompt
Scenario Generator	The controls available to researchers to tweak the behavior of the system (i.e. generate different conversation scenarios)
SwiftUI	Framework for creating interface with the Swift coding language
PR	Pull Request
CI	Continuous Integration
"App" or "The App"	The application we seek to develop for the Apple Vision Pro

Relevant Facts and Assumptions

- For the purposes of this project, we expect that the user/participant will have or will be provided with an Apple Vision Pro headset.
- We expect that the user/participant is physically able to use the hardware sufficiently (eyesight, hearing, cognition) or is provided with an aide.

Functional Requirements

#	P0-P3	Requirement
1.	P0	The application shall confirm research consent before any capture begins.
2.	P0	The application should provide session control features, namely *start, pause,* and *stop*.
3.	P0	The application, when directed, shall capture live conversational input and produce a text-formatted transcript of the dialogue.
4.	P0	The application shall detect indicators of low engagement, such as but not limited to prolonged silence, hesitation, muttering, and non-conversationally relevant speech.
5.	P0	Upon detection of low engagement, the application shall surface (with audio or visual elements) a prompt to help continue the conversation
6.	P0	The application, upon termination, shall provide a minimal post-session summary to verify capture

7.	P0	The application shall store session artifacts (transcription, processed audio-recordings, timestamps, prompts shown) locally for the purpose of research inspection, statistical analysis, and compliance
8.	P0	The application will support visionOS 26+
9.	P1	The application shall integrate with an LLM to generate real-time, context-aware conversational support prompts
10.	P1	The application shall be capable of using multiple different LLM APIs, including OpenAI's GPT-4/GPT-5, Google's Gemini, etc.
11.	P1	The application shall upload recorded session data, including transcripts, timestamps, and AI-prompts in a structured CSV or JSON format.
12.	P1	The front-end of the application shall be designed and optimized to reflect feedback from older-adults and prioritize accessibility and clarity.
13.	P1	The application shall classify different types of low engagement and adjust its behavior accordingly.
14.	P1	The application shall automatically anonymize (or pseudonymize) participant data prior to export or storage to ensure compliance with research ethics and data-protection regulations.
15.	P1	The application shall allow researchers to configure experimental scenarios by adjusting variables such as LLM used, AI-support levels, context window sizes and position, timing of prompt delivery, etc.
16.	P2	The application shall support downloading session data in standard research formats (e.g. CSV, XLSX) for external processing, with support for searching and filtering by parameters such as date, LLM used, etc.
17.	P2	The application shall compute and export metrics (such as average stall duration, response latency, and number of AI prompts per session) for statistical analysis, aggregated based on the session configurations.
18.	P2	Before a session begins, the user is brought through a short onboarding tutorial demonstrating what the prompts will look like.
19.	P3	The application will be thoroughly documented for both technical maintainers and end users (researchers and participants).
20.	P3	Use Apple's "Persona" system to gauge the user's emotional state and adjust prompts accordingly to guide the conversation to be more positive.
21.	P3	Conduct a brief cognitive assessment during the onboarding process.
22.	P3	Web-based dashboard to easily view and download both raw and aggregated data and statistics collected during research sessions.

Data and Metrics

Data Collection

The app will capture & store data relevant to monitoring conversational engagement and AI intervention during participant interactions.

Category	Data Type	Description
Audio	Raw Encrypted Microphone Input	Speech recognition & analysis. Ethics will determine persistence of data
Transcripts	Text from speech recognition software	Input for NLP & summarization LLMs as well as conversational analysis
Timestamps	Start/Stop times for pauses, responses & utterances	For calculating latency, pause duration, and exchange frequency
AI Interventions	Info on when & how prompts were provided	Tracking frequency, timing & types of LLM suggestions
Session Metadata	Session ID, participant aliases, consent flag	Traceable app sessions without compromising the privacy of the user
User Interaction Logs	Acceptance/rejection of AI suggestions, and gaze data that is possible to retrieve within the tightly secure Apple Vision Pro environment	Measures the response of the participant/user to the app interfaces

Metrics

The app will compute specific quantitative measures of the user's engagement within conversation and the LLMs interaction quality.

Metric	Formula / Basis	Purpose
Exchange Length	Average number of words per turn	Proxy for conversational richness.
Response Latency	Time between end of one speaker's utterance and start of the other's	Indicates conversational flow and engagement.
Pause Frequency	Number of pauses > x ms per minute	Identifies disruptions or hesitations.
AI Intervention Rate	Number of prompts per session or per minute	Quantifies system intrusiveness.
AI Suggestion Acceptance Rate	Accepted ÷ total suggestions	Evaluates perceived usefulness of AI help.
Session Duration	Total elapsed time per conversation	Tracks overall engagement and endurance.
Word Overlap / Topic Continuity	NLP similarity between adjacent utterances	Measures topical coherence of conversation.

Data Quality & Performance Metrics

Measuring internal performance will be tracked through the following.

System Metric	Target / Goal
Event Trigger Engine Accuracy	≥ 85% correct detection of pauses, loss of focus, or conversational breakdowns compared to annotated ground truth.
Speech Recognition Accuracy	≥ 90% transcription accuracy in quiet environments
Conversational Stall Rate	≥ 30% reduction in average stall duration compared to baseline (no AI support).
LLM Response Latency	≤ 5 seconds average from pause detection to displayed prompt.
Prompt Acceptance Rate	≥ 70% of AI prompts accepted or acted upon by users.
Suggestion Relevance Rating	≥ 4/5 average user evaluation post-session.
Anonymization Verification	100% of stored records pseudonymized; no raw identifiers retained.

Non-functional Requirements

Look and Feel Requirements

The mixed reality app should have a calm, clear, and easy-to-navigate interface that uses soft colors with good contrast and large, readable sans-serif text. Visuals and animations should feel natural and purposeful, maintaining a professional yet warm tone. The overall usability should focus on minimizing cognitive load through simple layouts, smooth interactions, and clear, friendly prompts that guide users without overwhelming them.

Usability and Humanity Requirements

The application should minimize cognitive load and make participation effortless for older adults and researchers. Interactions must feel natural, with simple voice or gesture commands and minimal steps required to complete tasks. The system should provide gentle feedback, maintain low mental demand, and encourage a sense of confidence and ease throughout the session. The overall tone and interaction style should feel polite, supportive, and human, helping older adults feel at ease while maintaining focus on meaningful conversation.

Performance and Speed Requirements

The app should run smoothly on Apple Vision Pro without noticeable lag or frame drops. Real-time processing is important since the system needs to listen, analyze, and respond during live conversations. Speech-to-text processing and AI-generated prompts should happen within a few seconds to keep interactions natural. Visual and audio cues should appear as quickly as possible when engagement drops, so the older adult can respond right away. The app should also manage resources efficiently. Logging and data collection should happen in the background without interrupting the user experience. Overall,

performance should feel fast and reliable, supporting real-time feedback without technical distractions.

Operational and Environmental Requirements

The operational and environmental requirements for the project ensure that the application functions reliably across different real-world research settings while maintaining user comfort and system stability. The product is expected to operate primarily indoors in quiet, well-lit environments such as research labs, clinics, or community centers, but the speech recognition systems should still perform effectively with moderate background noise. Overall, the app should deliver consistent performance, comfort, and reliability across various environments, ensuring researchers can confidently conduct studies without technical interruptions.

Security and Privacy

The system shall ensure that all conversational data and participant information are collected, processed, and stored in compliance with institutional research ethics and privacy regulations:

- **Data Encryption:** All raw audio, transcripts, and metadata shall be encrypted in transit and at rest using a standard at least as strong as AES-256.
- **Anonymization:** Any data collected will be tied only to random session IDs and never to an individual participant. Additionally, audio recordings will be processed to provide anonymity through random pitch shifting, and transcripts will have any names replaced with pseudonyms.
- **Limited Retention:** Any data collected will be annotated with a date such that data can be removed once it is no longer necessary for the study it was generated from.
- **Access Control:** Research data will be accessible only to authorized researchers through secure authentication and role-based access. No data will be shared externally without ethics board approval.
- **External Services:** Integration with remote APIs (e.g., GPT) shall minimize shared content to only the necessary text fragments, excluding identifiable or sensitive information as much as is possible with reasonable effort.

Risks and Issues

Risk/Issue	Description	Mitigation Strategy
Device Access Limitations	Limited access to Apple Vision Pro may slow development/testing	Schedule device time early; simulate with VisionOS simulator where possible

Unfamiliar Technology Stack	Learning VisionOS, SwiftUI, or other components may take longer than expected	Share learning, use tutorials, and ask supervisor for feedback regularly
Voice Recognition Accuracy	Misinterpretation due to varied speech or background noise	Alert users if the environment is too noisy, allow users to repeat themselves with a simple voice or button prompt, use AI models to improve accuracy over time by learning from each user's unique speech pattern
Misinterpreting Natural Pauses	The app may falsely detect normal thinking pauses as conversation breakdowns, triggering suggestions inappropriately	Use AI/LLM-based context analysis combined with predefined rule-based thresholds (e.g., pause duration, previous response type) to distinguish reflective pauses from true conversational stalls. Adapt models over time using real user interactions.
User Privacy Concerns	Logging even anonymized data may raise ethical/privacy concerns	Ensure de-identification, avoid assigning session data/transcripts with a specific participant identifier, avoid collecting PII, potentially do some voice processing to change the voice of participants when recording any audio
Ethics Approval	Formal ethics approval may take longer than expected, blocking real-participant testing.	Use simulated/scripted data first; finish consent/withdrawal flows early; supervisor pre-review; plan milestones that don't depend on human data.

User Documentation and Training

Comprehensive user documentation will be developed to support both researchers and older adult participants in the effective use of the system. This documentation will include a concise user manual outlining system setup, operation, data management procedures, and troubleshooting guidelines. Supplementary quick start guides and annotated interface diagrams will be provided for ease of use during study sessions. Training sessions should be conducted for researchers and facilitators prior to participant testing to ensure consistent and ethical operation of the system. Older adult participants should receive a brief, guided orientation emphasizing accessibility, comfort, and safety in using the mixed reality interface. All materials should prioritize clarity and simplicity to ensure accessibility for the older adults that will be using the system.

Project Development Plan

Team Communication Plan

Our team communicates primarily through Microsoft Teams and Discord. We hold a weekly meeting every Tuesday at 7:00 PM on Teams to discuss progress, blockers, and upcoming tasks. We also meet on Fridays at 2:00 PM with our supervisor, typically in person when hands-on interaction with the Apple Vision Pro device is required. For quick or ad-hoc updates, we rely on Discord, which enables fast, asynchronous communication. Team members unable to attend a meeting notify the group in advance via Discord and are brought up to speed through posted meeting minutes and assigned follow-ups.

Team Member Roles

In general, we expect that team members will work collaboratively on most activities. To assist in assigning tasks and effectively managing our knowledge base, however, each member will be assigned a main area of responsibility and the corresponding functional / non-functional requirements. They will then lead the group's efforts in that area and be the go-to person if any questions or issues arise regarding their section.

- **William Clubine**
  - Project Lead / Project Manager
    - Includes organizing and moderating team meetings, administering the GitHub organization and repository, monitoring deliverables' progress to ensure deadlines are met, leading communicating with the faculty supervisor, and assisting in the implementation and coordination of all other areas of the project as necessary
    - **Functional Requirements:** 19
  - Data Collection Lead
    - Includes all aspects of the project related to data collection, processing, storing, and analyzing, during and after test sessions
    - **Functional Requirements:** 6, 7, 11, 14, 16, 17, 22
- **Khach Kehyayan**
  - Meeting Scribe
    - Responsible for turning meeting transcripts into digestible minutes and uploading meeting summaries to the GitHub repository
  - VisionOS Co-Lead
    - Serves as a "resident expert" for everything related to the Apple Vision Pro's hardware, operating system (visionOS), and our app's frontend

        - **Functional Requirements:** 1, 2, 5, 6, 8, 12, 15, 18, 20, 21
- **Gregory Archer**
    - VisionOS Co-Lead
        - Shared responsibility with Khach Kehyayan
- **Nicholas Zajkeskovic**
    - Speech Recognition Lead
        - Responsible for everything to do with processing raw audio input into machine readable conversation transcripts
        - **Functional Requirements:** 3, 4, 7, 14
- **Jasiri Wa-Kyendo**
    - Prompt Generation Lead
        - Includes anything to do with generating the conversational prompts displayed to users when requested by the Event Trigger Engine
        - **Functional Requirements:** 5, 9, 10, 13, 15, 20
- **Mayowa Adesanya**
    - Event Trigger Engine Lead
        - Leads research and development of the Event Trigger Engine to intelligently detect when the user needs a conversational prompt
        - **Functional Requirements:** 4, 10, 13, 15, 20, 21

Workflow Plan

Source Code and Project Management

We will utilize GitHub as the central platform for source control and project management. All new work will be conducted in dedicated feature branches named using *hyphenated-lower-case* for clarity and consistency. Each branch will correspond to a GitHub issue, which will describe the related functional or non-functional requirement, user story, acceptance criteria, and responsible team member. These issues will also be our main tool for project management in combination with GitHub Projects to provide a kanban / sprint planning board. Before merging into the main branch, contributors must open a pull request (PR), obtain at least one peer review, and ensure that all continuous integration (CI) workflows and unit tests pass successfully. When a PR modifies code primarily written or maintained by another member, that member's review will be mandatory prior to approval as well.

Agile Methodology

The team will adopt an Agile development methodology, structured around two-week sprints. Sprint planning will take place during Tuesday meetings at the start of each new

sprint, while mid-sprint meetings will serve to assess progress and identify blockers. Sprint reviews will occur during Friday meetings with the faculty supervisor, where completed work will be demonstrated and feedback collected. Each sprint will have an assigned scrum master to lead these meetings, who will rotate over the course of the project to provide each member with scrum master experience, following the order that personnel are listed in the *Team Member Roles* section. This iterative approach will support continuous improvement and alignment with research and development goals.

Data Storage and Compute

All data collected from user testing will ultimately be stored in a cloud-based database, as described in the *Technology* section below. Prior to the implementation of the full data pipeline, any interim or one-off datasets will be stored and shared through a SharePoint site maintained by the project supervisor. The project is not expected to require heavy computation for training large models, as most processing will occur either on-device or through external AI providers (e.g., OpenAI). However, if additional infrastructure become necessary, compute resources will be provisioned using facilities provided by McMaster's Department of Computing and Software.

Performance Evaluation

To evaluate performance and project outcomes, the team will combine qualitative feedback from user and researcher testing with quantitative analytics derived from recorded interaction data. Python-based data analysis tools will be used to assess metrics such as model accuracy and system responsiveness. Any analytical scripts or generated reports will be version-controlled within a dedicated repository directory to ensure reproducibility and proper documentation of research results.

Documentations

We will co-author documents using Microsoft Office 365, share working files in Teams → Files (SharePoint), and commit finalized versions to GitHub under /docs to maintain version control and visibility for the instructors.

Proof of Concept Demonstration Plan

To demonstrate the potential of our project, we will present a working Vision Pro app that detects short live conversation, detects a pause, and automatically generates and displays an AI-powered conversational suggestion using an LLM.

Goals

1. **Prototype VisionOS Application:** A functioning Vision Pro app executable through Xcode on a VisionOS simulator or physical headset.
2. **Real-Time Speech Processing:** Continuous audio capture and transcription using Apple's Speech framework.
3. **Pause Detection:** Simple silence threshold-based detection (e.g., ≥ 1.0 s of no speech) indicating possible disengagement.
4. **Mixed Reality Visual Interface:** A minimalistic interface showing transcribed conversation and discreet AI suggestions as spatial UI elements.
5. **LLM Integration:** Connection to the LLM API to generate 1 short, context-aware conversational nudge based on received conversation history

Demonstration

A short 1-minute demo conversational scenario between 2 people showing the loop in action with printed logs and LLM response to the Apple Vision Pro UI demonstrated from the perspective of the apple vision pro user.

The example scenario will be guided by the following representation of a user experiencing conversational lapse:

User: *"...and then we... uh..."* **[silence for ~2 seconds]**

Expected AI response to Apple Vision Pro UI: *"You could ask them what happened next."; "Maybe mention how that experience made you feel."*

Technology

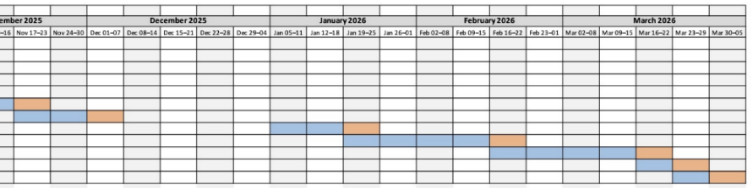
- **Frontend:**
  - **Swift**: Primary programming language for developing the VisionOS application.
  - **VisionOS SDK**: Provides tools and APIs for building immersive experiences on Apple Vision Pro.
  - **SwiftUI**: Declarative UI framework for creating responsive and adaptive 2D/3D interfaces.
  - **Next.js:** Web app framework for creating any web-based dashboards the project may require. This may not be required if web dashboards go out of scope.

- **Backend:**
  - **Go**: Fast, concurrent language for handling incoming requests, processing data, and calling AI provider APIs.
  - Alternatively, APIs may be used directly from the device, so no backend is needed.
- **Frameworks:**
  - **Apple Speech (AVAudioEngine / SFSpeechRecognizer)**: On-device speech recognition for capturing and analyzing participant conversations.
  - **XCTest**: Built-in testing framework for unit, UI, and performance testing.
- **Databases:**
  - **Supabase:** Cloud-based Postgres SQL database and object (file) storage provider, with the option to query directly from clients rather than through a backend server
  - Other options including alternative IaaS providers or on-device storage will be considered if a larger than expected amount of file storage is required
- **APIs:**
  - **LLM APIs (ChatGPT, Gemini, Llama, Grok)**: For AI-powered conversation analysis and context-aware suggestions.

Project Scheduling (Gantt chart)

As described in the *Workflow Plan*, this project follows an Agile development process organized into key academic milestones across the 2025–2026 academic year. The schedule integrates sprint-based work with the course deliverables, ensuring steady progress toward both technical completion and evaluation requirements. Each phase includes specific outcomes that align with project grading components.

Task	Start	End	Weight (%)
Team Formation & Project Selection	2025-09-09	2025-09-19	5
Project Confirmation & Refinement	2025-09-20	2025-09-26	0
Unified SRS & Project Plan	2025-09-27	2025-10-10	15
Sprint 1: Setup & Research Integration	2025-10-11	2025-10-24	0
Sprint 2: MVP Prototype Development	2025-10-25	2025-11-07	0
Proof of Concept Video	2025-11-08	2025-11-21	10
Sprint 3: AI Integration – Phase 1	2025-11-22	2025-12-06	0
Sprint 4: AI Integration – Phase 2	2026-01-06	2026-01-23	10
Sprint 5: Scenario Generator + Data Layer	2026-01-24	2026-02-21	0
Sprint 6: Research-Ready Build + Testing	2026-02-22	2026-03-21	0
Final Demonstration Video	2026-03-22	2026-03-27	10
Final Submission & Poster Presentation	2026-03-28	2026-04-04	40



Legend
Blue = Active during week
Orange = Milestone (task end)
Grey = Month band; light shading alternates weekly